

Assignment 2: Teaching Strategies for Active Learning Worksheet

Teaching Context

First, think about a specific teaching context—course, topic, student population, and so on—where you'd like to try out an active learning approach. Be sure to describe the context in your responses so that someone outside your discipline, or someone outside the future course where you imagine using this strategy, can understand. This kind of clear communication will help a peer reviewer, if you complete part 2 with a peer, and it will also strengthen your own ability to communicate your teaching knowledge and practice regardless of whether you complete part 2 with a peer or individually.

Astronomy 101, intro to astronomy. A general course, so science and non science majors. Probably a large class too. This particular lesson will be on the Earth and its seasons. The lesson will cover the Earth's orbit around the sun and its rotation. We will discuss what causes the seasons and why the seasons on the Northern hemisphere are the opposite of those on the Southern hemisphere.

Active Learning Strategy

Next, select a teaching practice that we've discussed which you would like to implement in a future teaching context. Some of the approaches we've discussed so far include:

- 1. Cooperative learning**
- 2. Peer instruction**
- 3. Lecturing**
- 4. Inquiry-based labs**
- 5. Problem-based learning**

I would incorporate several strategies for this lesson. We will use cooperative learning and peer instruction, we will end with a short lecture and I will give them a problem to write about before next class period.

I will start the lab by posing a series of questions about the seasons. The students will then pair up with each other and discuss why they think the seasons occur. I will then give them a clicker question asking them this. Then I will show them a picture of someplace in the Southern hemisphere during our summer and ask what season it looks like in the photo. Then the students will need to pair up again and discuss how the new information fits into their understanding of how the seasons work. After this I will pose another clicker question and then

have people explain the reasoning for their answers. I will finish with a short summary lecture to make sure all students have the correct understanding. Their homework for next week will be to answer the following question. Eva lives on another planet. Their planet rotates at a speed of XX and this image describes the orbit of the planet. Where Eva lives is marked by the red star, describe how they will experience seasons on their planet.

Principles of Learning in Your Strategy

Lastly, respond to the following questions, one for each of the Module 1 learning principles.

Prior Knowledge in Active Learning

When implementing the active learning strategy, what steps would you take to activate students' prior knowledge in ways that would help them learn effectively?

We already have knowledge about how we experience seasons, just from being alive. I will pose questions that tap into this knowledge by asking students to think about what they have experienced.

Knowledge Organization and Active Learning

How might you use the active learning strategy to help students develop more robust knowledge organizations in your domain?

They will have to connect the knowledge they have with their own experience of the seasons to the new information of how one might experience seasons on the Southern hemisphere. At the very end they will try to connect that to how one might experience seasons on another planet.

Iterative Practice and Feedback

What choices would you make when implementing the active learning strategy to make sure students had opportunities for practice and feedback?

They will get feedback from their peers and from me when we have full class discussions. They will be able to practice through multiple i-clicker questions and the homework.

The Affective Domain

When implementing the active learning strategy, what are two or three steps you could take to attend to the affective domain?

By having students work in pairs and give each other feedback students can learn to feel more comfortable with being incorrect and see it as a step in learning something new. Students will also be encouraged during our group discussion to work together to get at a common understanding of the seasons.